

Program: Diploma in Polymer Technology	
Course Code: 5091	Course Title: Tyre Technology
Semester: 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To understand the development, classifications and nomenclature of pneumatic tyres.
- To acquire knowledge on the different components of tyres, raw materials used and their design.
- To understand tyre building, curing and testing

Course Prerequisites:

Topic	Course code	Course Title	Semester
Engineering Drawing		Engineering Graphics	1
Materials Machinery		Natural Rubber Production	3
		Synthetic Rubbers	3
		Polymer Processing and Machinery	4
Formulation Compounding & Moulding		Technology of Elastomers	3
		Rubber Products Workshop	4
Testing		Polymer Science Lab	3
		Polymer Analysis Lab	3
		Rubber Technology Workshop	3
		Physical Testing Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Describe statistics, design and development of pneumatic tyres.	15	Understanding
CO2	Summarize design and production of pneumatic tyre and tube	15	Understanding
CO3	Explain design and production of cycle tyre, cycle tube, solid tyre and re-treading	14	Understanding
CO4	Generalize tyre and tube testing methods	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3				
CO3	3		3				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Durations (Hours)	Cognitive Level
CO1	Describe statistics, design and development of pneumatic tyres.		
M1.01	Describe history and development of pneumatic tyres.	3	Remembering
M1.02	Explain different components of tyre and their functions	5	Understanding
M1.03	Describe basic tyre constructions.	3	Understanding
M1.04	Outline composition of tyre.	4	Understanding
Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction			

- bias - radial - bias belted. Aspect ratio. Tyre size and dimension.

Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials

CO2	Summarize design and production of pneumatic tyre and tube		
M2.01	Comprehend design and production of tyre components	6	Understanding
M2.02	Compile adhesive treatments of tyre cords	2	Understanding
M2.03	Summarize tyre building and curing	4	Understanding
M2.04	Outline design and production of automobile inner tube and flap	3	Understanding
	Series Test I	1	

Content:

Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug.

Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords.

Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation.

Automobile inner tube -production - typical compound. Tyre flap - production - typical compound.

CO3	Explain design and production of cycle tyre, cycle tube, solid tyre and re-treading		
M3.01	Outline design and production of cycle tyre	5	Understanding
M3.02	Describe design and production of cycle tube	4	Understanding
M3.03	Explain design and production of solid tyre	3	Understanding
M3.04	Describe design of re-treading materials and re-treading process	2	Understanding

Content:

Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing.

Cycle tube: design and production.

Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre.

Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading.

CO4	Generalize tyre and tube testing methods.		
M4.01	Describe testing on tyre cords and compounds	4	Understanding
M4.02	Outline non-destructive and destructive testing of pneumatic tyres and tubes	4	Understanding
M4.03	Describe outdoor testing of pneumatic tyres	2	Understanding
M4.03	Summarize testing of cycle tyre and tubes	4	Understanding
	Series Test II	1	
Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests.			

Text /Reference:

T/R	BookTitle/Author
T1	Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber Company)
T2	Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer)
R2	Rubber Technology and Manufacture - C.M.Blow (Publisher - Butterworth Scientific)
R3	Vanderbilt Rubber Hand book - R.T Vanderbilt (Publisher - Vanderbilt Company Inc.)
R4	Rubber Technology and Manufacture - Published by SBP
R5	IS Specifications - Bureau of Indian Standards

Online resources:

Sl.No	Website Link
1	www.researchgate.net
2	www.scribd.com
3	www.iri.net.in